

Using Point-Slope Form

Sometimes you can’t find the y -intercept by looking at the graph. The y -intercept may be a fraction, or it may intersect at a point not shown on the graph. In this lesson, you will learn to use the **point-slope form**. Use this form when you don’t know the y -intercept.

The point-slope form appears on the GED math formula sheet.

point-slope form $y - y_1 = m(x - x_1)$, where m = slope and (x_1, y_1) is a point on the line

Example 1

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Write the equation of the line passing through $(5, 2)$ and $(7, -2)$.

Step 1

Use the slope formula to find the slope. Let $(5, 2)$ be (x_1, y_1) and $(7, -2)$ be (x_2, y_2) .

$$m = \frac{-2-2}{7-5} = \frac{-4}{2} = -2$$

Step 2

Substitute the values into point-slope form. You may use either point.

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 2 &= -2(x - 5) \end{aligned}$$

$$\begin{aligned} \text{or } y - (-2) &= -2(x - 7) \\ y + 2 &= -2x + 14 \\ y &= -2x + 14 - 2 \\ y &= -2x + 12 \end{aligned}$$

Step 3

Rewrite the equation in slope-intercept form.

$$\begin{aligned} y - 2 &= -2(x - 5) \\ y - 2 &= -2x + 10 \\ y &= -2x + 12 \end{aligned}$$

The equation of the line is $y = -2x + 12$.

You will also see linear equations written in **standard form**. The standard form of an equation is $Ax + By = C$, where A , B , and C are numbers. Standard form can be useful for finding the x - and y -intercepts, but it isn’t always the most helpful way to graph an equation. Use algebra to change standard form into slope-intercept form.

Example 2

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Graph the equation $2x - 3y = 9$.

Step 1

Write the equation in slope-intercept form by isolating y .

$$\begin{aligned} 2x - 3y &= 9 \\ -3y &= -2x + 9 \\ y &= \frac{2}{3}x - 3 \end{aligned}$$

$$\begin{aligned} \rightarrow y &= \frac{2}{3}x - \frac{9}{3} \\ y &= \frac{2}{3}x - 3 \end{aligned}$$

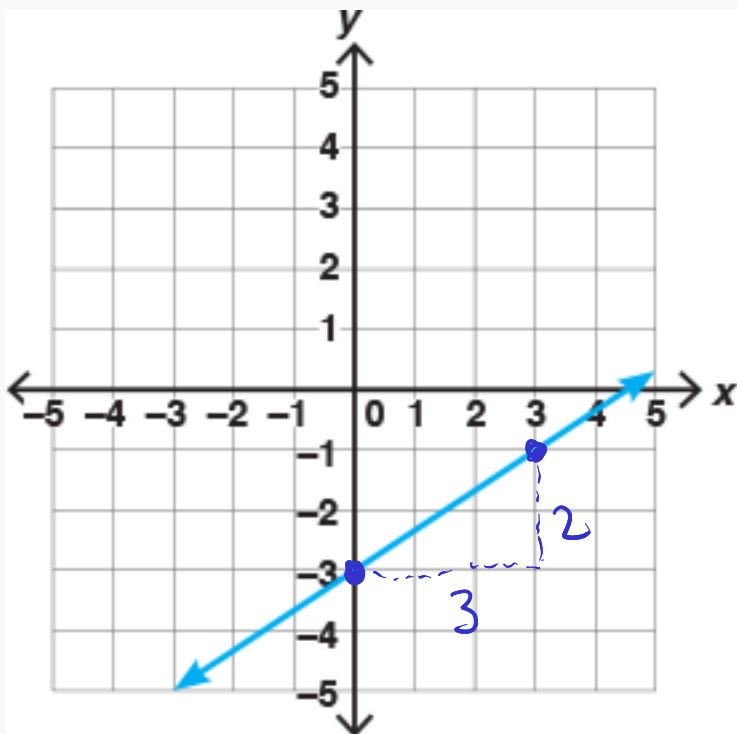
Step 2

Graph the equation using the slope and y -intercept.

$$y = \frac{2}{3}x - 3$$

The slope is $\frac{2}{3}$.

The y -intercept is $(0, -3)$.



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